Prob. 1: For the questions a, b, c, d & e consider the circuit shown in figure 1 and answer (use nodal analysis method):

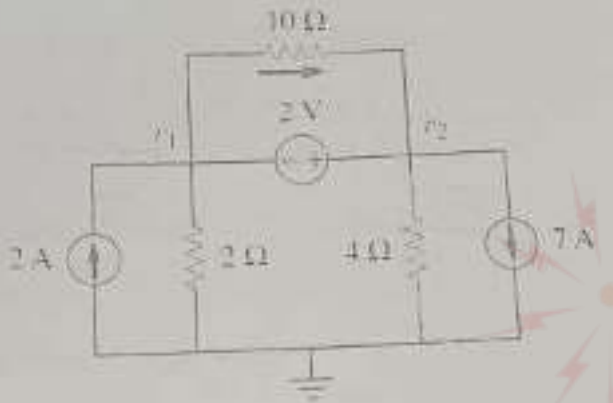


Fig.1

a. The value of V1 is: -7.333 [V] — 2 marks

b. The value of V2 is: -5.333 [V] — 2 marks

c. The value of the current (i) through 10Ω resistor is: -0.2 A — 1 mark

c. The power absorbed by 4 Ω resister is: 7.11 watt. — 1 mark

d. The power absorbed by 2 A current source is: 14.666 watt. — 1 mark

e. Write the required equations (do not simplify) — 2 marks

(2 V source + 10 Ω) Form supernode: $2 = \frac{V_1}{2} + \frac{V_2}{4} + 7$ — 1 mark

$-2 = V_1 - V_2$

Prob. 2: Consider the circuit shown in figure 2, and answer:

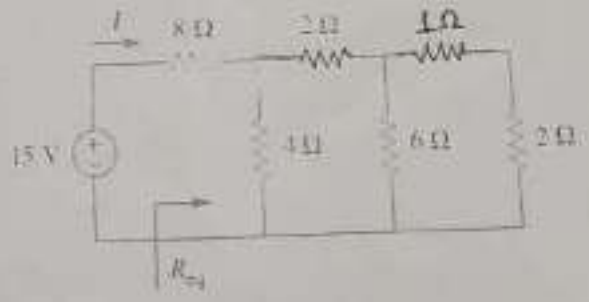


Fig.2

a. The equivalent resistance Req is: — 3 marks

$R_{eq} = 10\ \Omega$

. The current I is:

$I = \frac{15}{12} = 1.5\ A$ 3 marks

. The Power absorbed by 4Ω resistor is :

$(0.75)^2 \times 4 = 2.25$ watt 2 marks

b.3: a. The charge transferred through an element is given by $q(t)=2t^3+5t^2+6$ C. Calculate the rrent entering the elem nt at instant t =10 ms. 3 marks

$i = \frac{dq}{dt} = 6t^2 + 10t \Big|_{t=10ms} = 100.6\ mA$

$6[10\times10^{-3} + 10\times10^{-3}]A = 6\times100\times10^{-6} + 10\times10^{-3} = 6\times10^{-4} + 10^{-2}A = 0.6 + 100 = 100.6\ mA$

load draws 500W when connected to a 230 V line. Calculate the current and the energy consumed o hours in joules

1) $I = \frac{500}{230} = 2.174A$; 2) $W = P.t = 500 \times 2 \times 60 \times 60\ J$

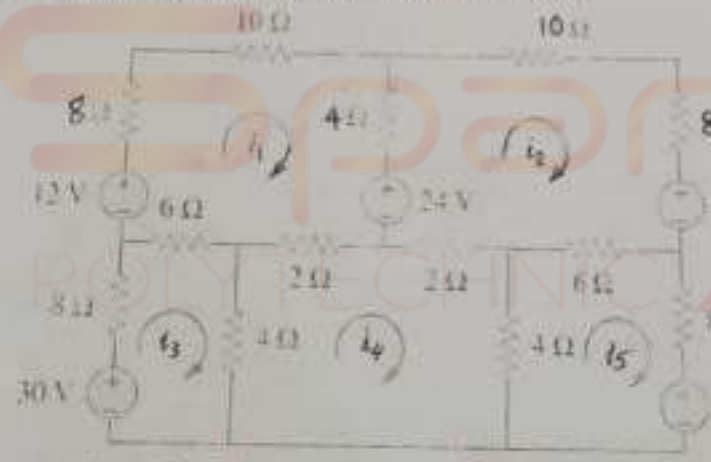
$= 1000 \times 3600\ J$

$= 3600\ kJ$

$= 3.6\ MJ$

3 marks

l: Refer to figure 3; write the mesh equations in matrix form:



[A] [i] = [v] -- Fig. 3 2 marks

$$\begin{bmatrix} 30 & -4 & -6 & -2 & 0 \\ -4 & 30 & 0 & -2 & -6 \\ -6 & 0 & 18 & -4 & 0 \\ -2 & -2 & -4 & 12 & -4 \\ 0 & -6 & 0 & -4 & 18 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \\ i_5 \end{bmatrix} = \begin{bmatrix} -12 \\ -16 \\ 30 \\ 0 \\ -32 \end{bmatrix}$$

4 marks